



Assessment Brief

Module title:	Programming for Data Analytics
Module code:	COM 7024
Assignment title:	Project
Assignment format:	MS Word / PDF
Word/time limit:	3000 words
File type	. docx / .pdf
Percentage of final grade	This assignment is worth 100% of your final grade for this module.
Submission deadline	See module iLearn page for date of submission
Grade release	You will normally receive your provisional grade and feedback within 20 working days of the submission deadline

Useful terms:

Learning outcomes (LOs)	The skills and knowledge that you should be able to show in your work.
Rubric/Marking Matrix	A set of rules or guidelines used to grade or assess work.

Task summary:

As part of the formal assessment for the programme you are required to submit a **Programming for Data Analytics** assignment. Please refer to your Student Handbook for full details of the programme assessment scheme and general information on preparing and submitting assignments. The assignment brief will specifically give details and instructions for the assignment. No examination, or details of, are included within this module.

Assignment instructions:

Description: The assignment is given as one extended task, unless multiple components have been given. Task 1 has been designed to check your understanding of Data Analytics programming methodologies for critical analysis, through a real-world mini–Data Analytics project. All of which will be written in Python. Plus, Task 1 allows you to demonstrate your reporting skills, comprehensively written in an easily understandable manner, based on your Data Analytics study.

A clear, concise analysis for the Task is to be given within the submission, complimented with screenshot evidence of all processes and results. You are to submit a single Word document for the task, and your code for Task 1 is to be included within an appendix so that it can be checked and verified it be working correctly. Your student ID number must be clearly defined in the uploaded file.

Assignment Task

Task 1 – Data Analytics in Python

Detail Task 1

Having secured a junior position at a German real estate company, the data analyst manager has requested that you to perform exploratory data analysis, using Python, on a Berlin housing data set, with focus being on the *price* and *total square foot* variables, plus a further study on whether there is a correlation between *living area* and *living square foot*. The purpose of the study is to determine basic analytics to help predict possible housing market trends in the future within Berlin. Therefore, you are to correctly import the dataset i.e. **BerlinHousing4049.csv** and perform an initial basic statistical test on all variables, once ensuring all data types are correct. Initial process now completed, perform all appropriate pre-processing methodologies, which you deem necessary and then perform a further basic statistical test to demonstrate improvements from the pre-processing stage.

Based upon the specific variables mentioned above, perform what you determine are appropriate investigations, with graphical representation, so that a meaningful study can be presented to your analysis manager. Full reasoning and justification is to be given for each screenshot section of code for all processes and chosen investigations.

Additionally, write a clear and concise report to your analysis manager defining your investigation. Your report will include all relevant analysis methodologies undertaken, full reasoning for your chosen analysis techniques and recommendations.

(100 Total Marks)

(LO's: 1, 2, 3 & 4)

(2000 Word equivalence for coded works)

(1000 Words Reflective report)

(Note: In this assignment, we will not select or apply any ML model, only apply preprocessing techniques including EDA, and write a reflective report based on the findings and at the end suggest some insightful recommendations that would help for further analysis and support in the decision-making process.)

End of Questions

[4049]

Learning outcomes (LO)

By completing this assessment, you will have shown and be assessed on **all** four of the learning outcomes:

1. Demonstrate a critical understanding of the concepts and principles of programming and programming paradigms. **[LO 1]**
2. Employ relevant tools, programmatic techniques, and features to prepare, manipulate, and analyse data. **[LO 2]**
3. Critically evaluate the preliminary results of manipulating and analysing data **[LO 3]**
4. Evaluate and apply digital tools and/or services while critically reflecting on opportunities for developing novel digital capabilities. Identify, select, plan for, use, modify, and evaluate digital applications and strategies to enhance the achievement of aims and desired outcomes. **[LO 4]**

Your marker will use a rubric/ marking matrix to grade your work, and you can access this by clicking on the submission portal.

Guidelines and policies

You can find links to more useful information about the assignment and university policies below.

Word/time limit policy

[Click here to view the Arden University word count/time limit policy](#)

Referencing guidelines

Please follow the referencing guidelines that are appropriate for your degree programme. If you are unsure which you should be using, please contact your module team.

[Click here for Harvard referencing guidelines](#)

Academic integrity and misconduct policy

[Click here to view Arden University's policy on academic integrity and misconduct](#)

Statement on use of artificial intelligence on assessment

[Click here to view Arden University's statement on the use of artificial intelligence on assessment](#)

Support information

[Click here to view guidance on how to apply for short-term extensions](#)

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